

Does a Prediction Market Beat the Weather Service?

Kalshi Temperature Ladders against NOAA's Own Probabilistic Guidance

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Abstract

Prediction market prices are read as probabilities, but they are rarely scored against a professional forecast of the same quantity, because for most events no such forecast exists. Daily maximum temperature is an exception. The National Weather Service publishes, through the National Blend of Models (the Blend in what follows, NBM in the tables), an operational probabilistic forecast of exactly the quantity Kalshi's temperature ladders settle on, at the same named stations, archived and free.

On 26,583 contracts across 6,023 city-days in eight US cities between August 2021 and November 2025, the market's price 24 hours before close is a better forecast than NOAA's guidance issued at the same lead. Brier scores are 0.1355 against 0.1564, a difference of 0.0209 with an event-clustered t of 22.2 and a bootstrap interval of 0.0191 to 0.0228. The market wins in all eight cities.

Two results give the finding its content. First, the market is not relaying the public bulletin. Regressing the outcome on both forecasts in log-odds, the market enters at 0.98 with a t of 42.1 while the Blend falls to 0.16; the market encompasses the guidance rather than repeating it. Second, a Murphy decomposition shows the advantage is not a trade of sharpness against calibration. The market is better on both terms: reliability 0.00027 against 0.00206, and resolution 0.03396 against 0.01521.

The mechanism is not the obvious one. The Blend's published spread is right on average: standardising the realised maximum by the Blend's own stated standard deviation gives a variance of 1.002 on 4,407 events, so the guidance is not underdispersed. What it cannot do is vary that spread with the day. Where the Blend states a standard deviation of one degree the realised dispersion is 1.44, and where it states four degrees the realised dispersion is 0.82. Its uncertainty is close to regime-invariant, so it fails to separate easy days from hard ones, and that is a resolution deficit rather than a calibration one.

1. Introduction

Whether prediction market prices should be read as probabilities is an old question, usually settled by comparing prices to realised frequencies. That tests internal consistency. It does not answer the question a policymaker or a forecaster would ask, which is whether the market knows more than the professional whose job it is to know.

Answering that requires a benchmark forecast of the same quantity, issued at the same time, in probabilistic form. Such benchmarks are rare in the settings prediction markets usually cover. Election markets have polls, which are not probability forecasts of the outcome. Sports markets have closing lines, which are themselves market prices. Macroeconomic markets have consensus surveys, which report point estimates.

Daily maximum temperature is the exception, and it is close to an ideal setting.

This question has been asked before, and the paper's contribution is narrower than the question. In April 2026 Interactive Brokers reported that forecasts from its own ForecastEx daily-high-temperature markets beat the National Weather Service's Localized Aviation MOS Program at the same stations, with a mean absolute error of 1.95°F against 2.61°F and the market ahead in twenty of twenty-three cities. That is this paper's headline, made first, by a market operator about its own venue. It also reports the two at near parity around twenty-four hours out, with the market pulling ahead as the target day progresses, which lands on precisely the horizon used here and is addressed in Section 7. Luo (2026) studies Kalshi weather markets as hedging instruments on 15,994 resolved contracts and reports, as this paper independently confirms, that out-of-sample recalibration does not improve the raw prices. Le (2026) covers 26,911 Kalshi weather markets in a broader calibration study.

What is new here is the treatment rather than the claim: a full ladder distribution scored against a probabilistic benchmark with a proper scoring rule rather than the absolute error of a point forecast, inference clustered on the shared weather outcome, a Murphy decomposition that identifies which component of forecast quality differs, and an encompassing regression that separates knowing something from relaying the public bulletin. No prior study scores a prediction market against the Blend, or against any NWS probabilistic guidance, with a proper scoring rule.

The benchmark exists and is public. The National Blend of Models is the National Weather Service's operational statistical guidance. Its station bulletins publish, for each station and each lead time, the mean, the standard deviation and the tenth, twenty-fifth, fiftieth, seventy-fifth and ninetieth percentiles of daily maximum temperature. That is a full quantile forecast of exactly the settled quantity. It is free, requires no account, and is archived from May 2020, before the Kalshi sample begins.

The contracts are ladders. Kalshi lists daily maximum temperature as disjoint two-degree buckets with two open tails, exhaustive and mutually exclusive, so a ladder is a discrete distribution and can be compared with one.

Settlement is exogenous and named. Each series settles on a specific NWS station's Daily Climate Report: Central Park for New York, Midway for Chicago, Bergstrom for Austin. Outcomes are objectively measured, unmanipulable, and not subject to insider information in any ordinary sense.

And the comparison is fair in a way most are not. Both forecasts are scored on the same events, with the same rule, at the same lead.

This paper reports that the market wins, that it wins for a reason that survives controlling for the benchmark, and that the reason is visible in the shape of each forecast's errors.

What is not claimed. This is not a demonstration that crowds beat meteorologists. The Blend is automated statistical guidance, not the official NWS forecast, which is issued by human forecasters who themselves adjust it. Kalshi's weather traders can read the Blend and some demonstrably do, so the paper's question is deliberately narrower and is stated in Section 5: does the market add information beyond the public guidance? It does. Nor is any trading claim made: the market's edge is measured in Brier score, not in returns net of fees and spread.

2. Data

Markets. Every resolved Kalshi daily maximum temperature contract for the eight cities the exchange has listed: New York, Chicago, Austin, Miami, Denver, Philadelphia, Los Angeles and Houston. Requiring at least three trades gives **27,654 contracts across 6,068 city-days**, closing between 6 August 2021 and 22 November 2025.

Contract structure changed over the sample and this matters for one comparison. In 2021 and 2022 Kalshi listed these as single threshold contracts, "91 degrees or higher". The exhaustive bucket ladders arrived later: the share of city-days carrying a genuine ladder runs 0.00 in 2021, 0.48 in 2022, 0.70 in 2023, 0.87 in 2024 and 0.99 in 2025, and 0.79 across the sample. The head-to-head comparison is valid for either structure, since it scores contracts individually; anything relying on the ladder adding up to one is reported separately on the 4,800 ladder events.

The benchmark. the Blend probabilistic station bulletins, cycle 01Z, from the NOAA Open Data bucket on AWS. The 01Z bulletin issued on date D carries, in its first column, the maximum temperature valid 00Z on D+1, which is the daytime maximum of local day D. That is a lead of **23 hours**, and it is the column used throughout.

Each Kalshi bucket label is converted to continuous bounds on the reported integer maximum, so "70 to 71" becomes the interval from 69.5 to 71.5, and "86 or above" becomes 85.5 to infinity. The Blend's probability for a bucket is the mass its forecast distribution places between those bounds.

Constructing the Blend's probability, and giving the benchmark its best case. Two constructions were tested: a normal from the published mean and standard deviation, and a piecewise-linear interpolation through the five published percentiles with normal tails. The normal scores better **for the Blend**, 0.1564 against 0.1575, and is therefore used as the primary. Widening the Blend's standard deviation was also tested, at 1.25, 1.5 and 2.0 times, and makes its Brier score worse at every setting. The benchmark is judged on its own best showing.

The horizon, and why it is not the last trade. Kalshi keeps these markets open until 23:59 local time while the day's maximum is physically realised by mid-afternoon. The last traded price is therefore not a forecast but largely a settled position: its Brier score is 0.01553, against 0.13552 for the price 24 hours before close. Scoring the last trade would be scoring hindsight, and would flatter the market by an enormous margin for no reason. **Every comparison in this paper uses the price 24 hours before close**, matched to the 01Z the Blend cycle.

3. The market is the better forecast

Table 1. Brier scores.

Forecast	Brier	contracts	events
Kalshi, 24 hours before close	0.13552	26,583	6,023
NOAA National Blend of Models	0.15642	26,583	6,023
Naive baseline, one over legs	0.16674	26,583	6,023
NBM, renormalised within ladder	0.14856	23,670	4,800

The first three rows are the same contracts. The renormalised row is not, and is reported separately: renormalising requires a ladder whose buckets tile, which excludes the single-threshold contracts of the early sample.

Both forecasts beat the naive baseline that knows only the ladder's structure, but by very different margins: the market's skill score is 0.187, the Blend's is 0.062.

Table 2. Head-to-head, standard errors clustered on city-day.

Comparison	difference	s.e.	t	95% interval	contracts	events
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NBM minus market	0.02090	0.00094	22.2	[0.0191, 0.0228]	26,583	6,023
NBM renormalised minus market	0.01834	0.00096	19.0	[0.0165, 0.0202]	23,670	4,800

Clustering is on the city-day because the legs of one ladder share a single weather outcome and cannot be treated as independent.

Table 3. By city. The market wins in all eight.

City	market	NBM	difference	t	events
New York	0.14507	0.16746	0.02239	11.3	1,550
Chicago	0.14051	0.17408	0.03357	15.1	1,536
Miami	0.13472	0.14118	0.00647	2.8	907
Austin	0.13529	0.14996	0.01466	7.7	903
Denver	0.11828	0.14217	0.02389	7.7	368
Philadelphia	0.13053	0.14069	0.01016	3.1	367
Los Angeles	0.11424	0.14071	0.02647	7.3	322
Houston	0.13076	0.14427	0.01351	1.5	70

Restricting to the 4,800 genuine ladder events leaves the result intact, with a t of 19.1.

4. Why: reliability and resolution

A Brier score answers which forecast is better. Murphy's decomposition answers what it is better at:

$$\text{Brier} = \text{reliability} - \text{resolution} + \text{uncertainty}$$

Reliability penalises probabilities that do not mean what they say and is zero for a perfectly calibrated forecast. Resolution rewards separating high-outcome cases from low-outcome ones, and is what a forecast earns by knowing something. Uncertainty belongs to the events rather than to either forecaster and is identical for both.

Table 4. Murphy decomposition, twelve bins.

Forecast	Brier	reliability	resolution	uncertainty
Kalshi, 24h	0.13552	0.00027	0.03396	0.16981
NBM	0.15642	0.00206	0.01521	0.16981

The market is better on both terms. Its reliability penalty is smaller by a factor of nearly eight, and its resolution is larger by a factor of more than two. This is not a forecast trading sharpness for calibration; it is better at both jobs.

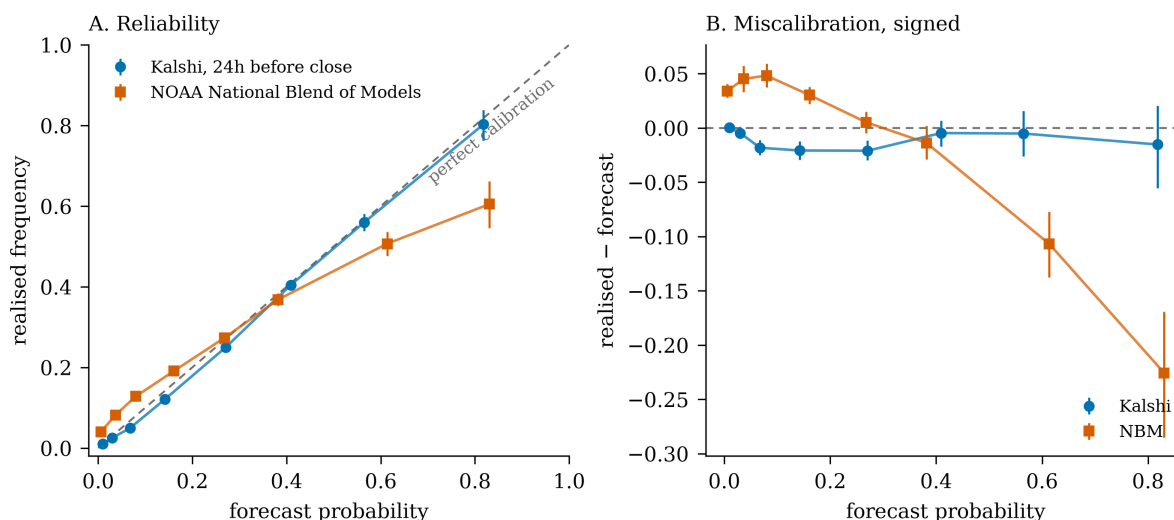


Figure 1. Reliability of both forecasts on the same 26,583 contracts. Panel A plots realised frequency against forecast probability with the 45-degree line of perfect calibration; Panel B plots the signed miscalibration, realised minus forecast. Bars are 95 per cent intervals from a bootstrap resampling city-days. The market tracks the diagonal within about two points everywhere. The Blend lies above the diagonal at low probabilities and falls well below it at high ones, the signature of an overconfident forecast: too much probability on the likely bucket and too little in the tails.

Figure 1 shows the shape of each forecast's error. The Blend's signed error runs from about +0.05 where it forecasts small probabilities to -0.22 where it forecasts large ones: a bucket the Blend calls 80 per cent likely happens about 60 per cent of the time. The market's errors are mildly negative in the low-probability range, about two points, which is the favourite-longshot bias familiar from betting markets, and negligible elsewhere.

The natural explanation of the Blend's shape is wrong, and the data say so. An underdispersed forecast, one whose spread is too narrow, would produce exactly that pattern, and statistical guidance is often underdispersed. Here it is not. Take the winning bucket on each of the 4,407 events where one leg resolved yes, put the realised maximum at its midpoint, and standardise by the Blend's own published standard deviation. The result has mean 0.056 and **variance 1.002**. On average the Blend's stated uncertainty is exactly right. Consistent with that, rescaling its standard deviation minimises the Brier score at 1.10 times, essentially where it already is, and widening it further makes the Blend worse.

Table 5. Realised dispersion, conditional on the uncertainty the Blend claims.

NBM states	realised sd of z	events
1°F	1.44	544
2°F	0.97	2,293
3°F	0.87	1,116
4°F	0.82	344

The spread is right on average and wrong everywhere in particular. The Blend is too confident on the days it calls certain and too diffuse on the days it calls uncertain: its uncertainty barely moves with the weather. A forecast whose stated uncertainty does not vary with the situation cannot separate easy days from hard ones, and that is precisely the resolution term in Table 4, where the Blend scores 0.01521 against the market's 0.03396. Its reliability penalty, 0.00206, accounts for under a tenth of the 0.021 gap. The market's advantage is that it knows

when it knows.

5. Does the market add information, or relay it?

The result so far could be explained without any market intelligence: if traders read the Blend and correct its known overconfidence, the market would beat it while contributing nothing of its own. This is not a hypothetical, because the Blend is free and Kalshi's weather traders demonstrably use it.

The forecast-encompassing test separates the two. Put both forecasts on the right-hand side in log-odds,

$$\text{logit } P(\text{yes}) = a + b_1 \cdot \text{logit}(\text{market}) + b_2 \cdot \text{logit}(\text{NBM}),$$

with standard errors clustered on the city-day. If the market is only relaying the Blend, b_1 is zero once the Blend is included.

Table 6. Forecast encompassing.

	coefficient	s.e.	t
Market, 24h	0.9806	0.0233	42.1
NBM	0.1575	0.0167	9.4
<i>Market alone, slope</i>	1.0755	0.0208	<i>3.6 against 1</i>
<i>NBM alone, slope</i>	0.5853	0.0191	<i>-21.7 against 1</i>

The market enters at essentially one and the Blend collapses to 0.16. The market encompasses the guidance: almost all of the information in the Blend is already in the price, and the price carries a great deal that the Blend does not. The Blend is not worthless, its coefficient is positive and significant, but it is a minor addition to the market rather than the other way round.

The single-forecast slopes make the same point differently. A calibrated forecast has a slope of one. The market's is 1.076, mildly underconfident. The Blend's is 0.585, decisively overconfident, and the gap from one is more than twenty standard errors.

6. Robustness

The result was tested against every cut that could plausibly conceal it, including the one that has broken adjacent work: a finding that exists in only part of the sample.

Table 7. The market's advantage, by subsample. Every cell is a separate head-to-head with errors clustered on the city-day.

Cut	<i>t</i> against NBM
Calendar year 2021 / 2022 / 2023 / 2024 / 2025	3.7 / 12.1 / 7.2 / 7.2 / 16.5
Season, winter / spring / summer / autumn	12.5 / 8.6 / 11.3 / 12.1
NBM's own stated uncertainty, low / mid / high	16.7 / 11.1 / 9.7
Day difficulty, hardest / middle / easiest	16.9 / 10.1 / 11.9
City, all eight	1.5 to 15.1
Genuine ladders only, 4,800 events	19.1

The advantage is present in every year, every season, every city, at every level of the benchmark's own uncertainty, and on both easy and hard days. Both forecasts improve over the sample, the market's Brier falling from 0.203 to 0.117 and the Blend's from 0.252 to 0.137, but the gap does not close.

Selection. Scoring requires a trade at least 24 hours before close, which retains 96.1 per cent of the panel. The 1,071 dropped contracts are thin, a median of 10 trades against 66 and a median volume of \$654 against \$3,048, and resolve yes less often, 0.122 against 0.217, so they are disproportionately far-out-of-the-money legs. The restriction is small and its direction is disclosed rather than assumed harmless.

Benchmark construction. Reported in Section 2: the Blend is scored on the better of two constructions of its bucket probability, and widening its spread was tested at three settings and makes it worse.

7. What this does and does not establish

It does not show that crowds beat meteorologists. the Blend is automated statistical guidance. The official NWS forecast is issued by human forecasters who adjust that guidance, and it is not scored here. The right reading is that a market beats the automated public product, which is the product most downstream users actually consume.

It does not show the market is unbiased. It carries a small favourite-longshot bias of about two points at low probabilities, in the same direction the betting literature documents.

And its calibration slope is not a single number. Le (2026) reports Kalshi weather slopes below one, meaning prices too extreme, where this paper reports 1.076, meaning prices not extreme enough. Both are right about different objects and the difference is the weighting. Using one price per contract, as here, the slope is 1.076. Weighting the same contracts by their trade count, as pooling trades does, it is **0.727**, inside the range Le reports. Heavily traded legs sit near the money and are priced too extreme; the many thin legs in the tails are priced not extreme enough. Which number describes "the market" depends on whether one is asking about contracts or about dollars, and neither paper is in error.

It does not support a trading claim. The advantage is measured in Brier score. Nothing here nets out fees, spread, or the cost of holding a position for a day, and the buckets where the market's edge is largest are not necessarily the liquid ones.

The near-parity finding, addressed. Interactive Brokers reports that at roughly twenty-four hours out its market and NWS guidance are close to level, with the market's edge opening up during the target day. This paper finds a clear gap at twenty-four hours. Three differences are candidates and this paper cannot adjudicate among them: the venues differ, ForecastEx against Kalshi; the benchmarks differ, LAMP against the Blend; and above all the scoring differs, the absolute error of a point forecast against a proper scoring rule applied to a full distribution. A market can be no better than a model at predicting the single most likely value while being substantially better at saying how likely it is, and Table 4 is consistent with exactly that: the market's advantage is concentrated in resolution rather than in locating the central estimate. The two results are not necessarily in conflict, but the reader should know the earlier one exists.

It is not a statement about all horizons. At six hours before close the market's Brier falls to 0.0266 and at the last trade to 0.0155, but by then the day's maximum is largely realised and the comparison is meaningless. The 24-hour horizon was chosen because it is the shortest at which the market is still forecasting rather than recording.

What it does establish is that on a settled, objectively measured quantity, with a professional probabilistic benchmark published by the agency responsible for it, a regulated retail prediction market produced the better forecast on 6,023 consecutive city-days across eight cities, and did so by being both better calibrated and more informative.

8. Conclusion

Kalshi's temperature ladders are better forecasts of tomorrow's maximum temperature than the National Weather Service's own operational probabilistic guidance, at the same station and the same lead. The gap is 0.021 in Brier score with a t of 22.2, holds in all eight cities, and survives restricting to genuine ladders.

The market wins on both components of the score, and it wins for a reason that is not arbitrage on the benchmark: controlling for the Blend, the market's coefficient is 0.98 and the benchmark's is 0.16. The Blend's weakness is a specific and familiar one, underdispersion, and the market does not share it.

Data, provenance and replication

Market data are the public Kalshi market and trade records, read from a local parquet snapshot taken 23 November 2025. Benchmark data are NOAA National Blend of Models probabilistic station bulletins, cycle 01Z, from the public AWS Open Data bucket `noaa-nbm-grib2-pds`, retrieved for all 1,569 dates on which a temperature event resolved, with no missing dates. Station assignments follow Kalshi's own published contract terms.

Every estimate is produced by deterministic Python from those two sources in two modules, `pipeline/fetch\_nbm.py` and `pipeline/weather\_lab.py`, whose output is written to a machine-readable record. Language models were used to draft prose, to search the literature and to attack the argument; they computed nothing.

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